

Care Access Navigator

Cloud Security 1 — Criterion 2 Completion Report

Cloud Architecture Design & Security Patterns

This document is the completion report for CloudSec1 Criterion 2, built against the Care Access Navigator (HRMC) capstone's Azure environment (resource group rg-cloudsec1-can, East US). It covers the three-tier VNet build, the two implemented security design patterns (Bastion host and zero-trust NSG micro-segmentation), and the architecture diagram, with screenshots and compliance mapping tying each rubric line item to a working configuration.

1. Purpose & Scope

Purpose. Criterion 2 asks for hands-on proof that a cloud environment can be architected with defensible network security patterns, not just described on paper. This report documents the actual build: a three-tier Azure VNet, two implemented security patterns, and a private compute resource proven unreachable from the public internet.

Scope. Everything here was built inside a single Azure subscription's East US region, under the resource group rg-cloudsec1-can established in Criterion 1. It extends Criterion 1's Shared Responsibility framing directly — Criterion 1 identified the VM/container layer as IaaS (customer-managed network configuration, OS, and access control); this report is that customer-managed layer, built out.

Where this fits in the capstone timeline. Criterion 2's network architecture work sits alongside the capstone's own Week 2–3 topology and IAM design, and feeds forward into Week 9 (Controls & Compliance Mapping), where these NSG and Bastion controls will be formally entered into the project's HIPAA/NIST control matrix. As with the Cloud Security 1 track generally, this work is running ahead of the capstone's own week-by-week pace — noted here rather than assumed.

2. Status Summary

Rubric requirement	Status	Primary evidence
Three-tier architecture diagram (public/private subnets, security groups, NACL-equivalent)	Complete	Section 3 — Figure 1
Hands-on VNet configuration with subnet segmentation	Complete	Section 4
Security pattern 1 — Bastion host	Complete	Section 5
Security pattern 2 — Zero-trust NSG micro-segmentation	Complete	Section 6
Private resource with no public IP (proof pattern works end-to-end)	Complete (adapted — see Section 7 note)	Section 7

3. Architecture Overview

Care Access Navigator's on-premises design uses six VLANs (10.10.x.x). For this criterion, that topology is lifted into a single Azure VNet, can-vnet-cloudsec1 (10.20.0.0/16) — a deliberately distinct addressing scheme from the on-prem VLANs, kept in the same numbering style so the two environments read as one consistent portfolio.

In plain terms — why one VNet instead of two

The original plan paired Bastion with a true hub-and-spoke design (two peered VNets). Once built, the VNet turned out to be provisioned as a single 10.20.0.0/16 block rather than two separate address spaces, so a second peered VNet would have overlapped it. Rather than tear down and rebuild, the design was adapted to a single VNet carrying both patterns: Bastion for controlled remote access, and NSG-enforced zero-trust segmentation between tiers. Both patterns satisfy the rubric's "two security design patterns" requirement; only the specific pairing changed from what was originally scoped.

Figure 1 — Three-Tier Architecture Diagram

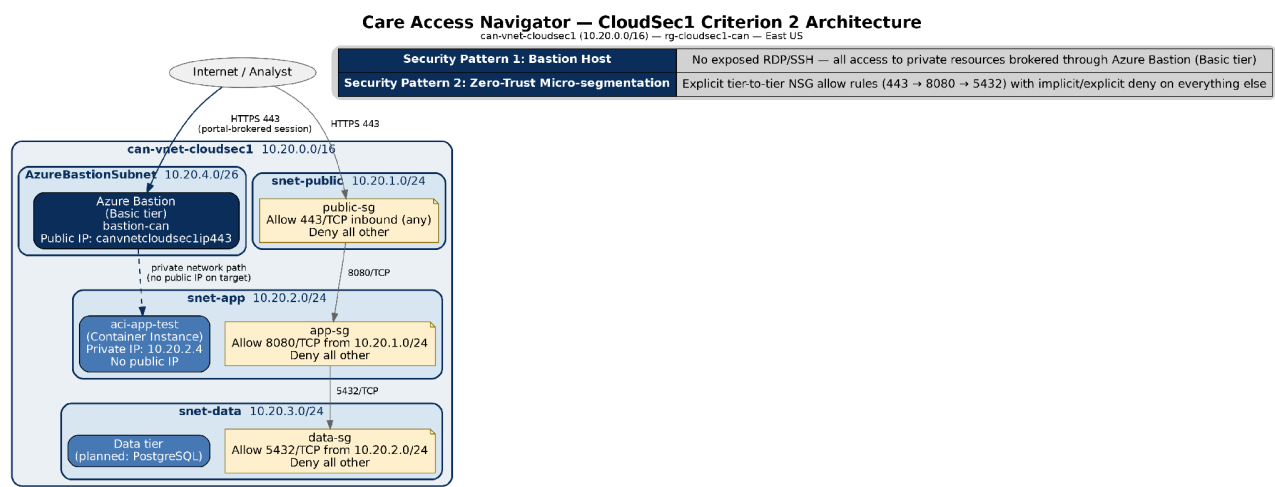


Figure 1. can-vnet-cloudsec1 (10.20.0.0/16): Bastion subnet, public/app/data tiers, NSG allow-chain (443 → 8080 → 5432), and the private compute resource with no public IP.

4. Hands-On VNet Configuration & Subnet Segmentation

can-vnet-cloudsec1 was built with four subnets: three application tiers plus a dedicated Bastion subnet, each carrying its own network security group.

All services > rg-cloudsec1-can

 **can-vnet-cloudsec1** ☆ ...

Virtual network

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

Move Delete Refresh Give feedback

Essentials [JSON View](#)

Resource group ([move](#))
[rg-cloudsec1-can](#)

Location ([move](#))
East US

Subscription ([move](#))
[Azure subscription 1](#)

Subscription ID
9a8c964b-cf59-4657-bff2-5f84fb7b0709

Address space
[10.20.0.0/16](#)

Advertised gateway prefixes
-

Subnets
[3 subnets](#)

DNS servers
[Azure provided DNS service](#)

BGP community string
[Configure](#)

Virtual network ID
658b9637-9738-4ee1-b3fc-5fbbb99c01e2

⌘ Add or remove favorites by pressing Cmd+Shift+F

Figure 2. can-vnet-cloudsec1 — address space 10.20.0.0/16, resource group rg-cloudsec1-can, East US.

can-vnet-cloudsec1 | Subnets

Virtual network

Search

+ Subnet Refresh Manage users Delete Export to CSV

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Address space
Connected devices
Subnets

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

☐	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table		
☐	snet-public	10.20.1.0/24	-	251	-	public-sg	-		
☐	snet-app	10.20.2.0/24	-	251	-	app-sg	-		
☐	snet-data	10.20.3.0/24	-	251	-	data-sg	-		

Figure 3. Subnet segmentation: *snet-public* (10.20.1.0/24 → *public-sg*), *snet-app* (10.20.2.0/24 → *app-sg*), *snet-data* (10.20.3.0/24 → *data-sg*).

Subnet purpose ⓘ Azure Bastion

Name * ⓘ AzureBastionSubnet

IPv4

Include an IPv4 address space ☒

IPv4 address range ⓘ 10.20.0.0/16
10.20.0.0 - 10.20.255.255

Starting address * ⓘ 10.20.4.0/26

Size ⓘ /26 (64 addresses)

Subnet address range ⓘ 10.20.4.0 - 10.20.4.63

IPv6

Figure 4. AzureBastionSubnet (10.20.4.0/26) added — the fixed subnet name Azure requires for Bastion deployment.

5. Security Pattern 1 — Bastion Host

Azure Bastion gives the environment a single controlled entry point for remote access, so private-tier resources never need a public IP or an exposed RDP/SSH port. Deployed at Basic tier (sufficient for this environment; Standard/Premium add scaling and auditing features not needed here) into AzureBastionSubnet, with Availability zone set to None and using the existing public IP canvnetcloudsec1ip443.

In plain terms — obstacle encountered — wrong tier on first deploy

The first Bastion deployment silently defaulted to Premium tier instead of Basic, and Azure does not allow downgrading a live Bastion instance in place ("Downgrading Bastion is not permitted"). The fix was to delete the Premium instance and redeploy from scratch through the resource group's Create flow (which surfaces the Tier selector explicitly) rather than the subnet-purpose shortcut. The redeploy also hit a second issue — the default Availability zone setting ("Zones 1,2,3") is incompatible with Basic tier and produced a misleading subnet-size validation error; setting the zone to None resolved it. Both are documented here as genuine configuration obstacles, not smoothed over.

can-vnet-cloudsec1-bastion | Configuration ☆ ... ✕

Bastion

◇ <<

-  Overview
-  Activity log
-  Access control (IAM)
-  Tags
-  Diagnose and solve problems
-  Resource visualizer
- ▼ Settings
 -  Sessions
 -  **Configuration**
 -  Identity
 -  Properties
 -  Locks
- > Monitoring
- > Automation
- > Help

RDP/SSH connectivity to one VM at a time for Dev/Test users with minimal workloads.

☒ **Basic**
The Basic tier enables dedicated RDP/SSH connectivity to multiple virtual machines in local or peered VNETs with support for NSGs.

☐ **Standard**
The Standard tier enables additional features, expanded connectivity options, and host scaling for heavier workloads.

☐ **Premium**
The Premium tier enables advanced auditing, security, and monitoring to secure sensitive workloads end-to-end.

✕ **Downgrading Bastion is not permitted**

Apply

Discard Changes

Figure 5. Premium-tier Bastion — downgrade to Basic blocked by Azure; deletion and redeploy required.

Home

Create a Bastion



Name *	can-vnet-cloudsec1-bastion ✓
Region *	East US ▼
Availability zone ⓘ	None ▼
Tier * ⓘ	Basic ▼
Instance count ⓘ	☐ 1 ☒ 2
Virtual Network ⓘ	can-vnet-cloudsec1 (rg-cloudsec1-can) ▼ Edit virtual network
Subnet ⓘ	AzureBastionSubnet ▼ Edit subnet 10.20.4.0 - 10.20.4.63 (64 addresses)
Configure IP Address	
IP Address ⓘ	☒ Public IP addresses ☐ Private IP address
Public IPv4 address	
Public IP address * ⓘ	☐ Create new ☒ Use existing
Choose public IP address *	canvnetcloudsec1ip443 (20.228.134.92) ▼

Figure 6. Redeploy with Availability zone: None + Tier: Basic — validation error cleared.

Create a Bastion ...

✔ Validation passed

- Basics
- Advanced
- Tags
- Review + create**

Summary

Basics

Name	can-vnet-cloudsec1-bastion
Subscription	Azure subscription 1
Resource group	rg-cloudsec1-can
Region	East US
Availability zone	None
Virtual network	can-vnet-cloudsec1 (rg-cloudsec1-can)
Tier	Basic
Subnets	AzureBastionSubnet
Public IPv4 address	canvnetcloudsec1ip443
Instance count	2
Copy and paste	Enabled
IP-based connection	Disabled
Kerberos authentication	Disabled
Shareable Link	Disabled
Native client support	Disabled
Session recording	Disabled
Private Bastion	Disabled

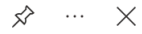
Figure 7. Review + create — validation passed, Tier: Basic confirmed before deployment.

Home



Microsoft.BastionHost-20260924154901 | Over...

Deployment



Delete



Cancel



Redeploy



Download



Refresh



Overview



Inputs



Outputs



Template



Your deployment is complete



Deployment name : Microsoft.BastionHost-20260924154901

Subscription : [Azure subscription 1](#)

Resource group : [rg-cloudsec1-can](#)

Start time : 9/24/2026, 3:49:03 PM

Correlation ID : 1569b1f0-fc91-4c31-b13c-2d29b7f8dbe1



Deployment details



Next steps

[Go to resource](#)

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Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

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Figure 8. Bastion deployment complete — Basic tier, can-vnet-cloudsec1-bastion, rg-cloudsec1-can.

6. Security Pattern 2 — Zero-Trust NSG Micro-Segmentation

Each tier's NSG was tightened from Azure's default (allow all VNet-internal traffic) to an explicit tier-to-tier allow chain: public-sg allows 443 inbound from any source; app-sg allows 8080 only from the public-tier subnet (10.20.1.0/24); data-sg allows 5432 (PostgreSQL) only from the app-tier subnet (10.20.2.0/24). Traffic outside this chain is denied by Azure's default DenyAllInBound rule (priority 65500) on every NSG. This is the "never trust, always verify between segments" principle applied directly to the network layer: no tier can reach another except through the one explicitly allowed path, regardless of being on the same internal VNet.

app-sg

Network foundation | Network security g... <<

app-sg | Inbound security rules ☆ ...

Network security group

Search << + Add Hide default rules Refresh Delete Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Inbound security rules

Outbound security rules

Network interfaces

Subnets

Properties

Locks

Monitoring

Automation

Help

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. [Learn more](#)

Filter by name

Port == all Protocol == all Source == all Destination == all

Action == all

Priority ↑↓	Name ↑↓	Port ↑↓	Protocol ↑↓
☐ 100	Allow-From-Public-Su...	Any	Any
☐ 65000	AllowVnetInBound	Any	Any
☐ 65001	AllowAzureLoadBalan...	Any	Any
☐ 65500	DenyAllInBound	Any	Any

Showing 1 - 3 of 3. Display auto

Figure 9. app-sg before: existing Allow-From-Public-Subnet rule at priority 100, Port/Protocol = Any/Any — too permissive for the intended pattern.



Allow-From-Public-Subnet

app-sg



Source ⓘ

IP Addresses



Source IP addresses/CIDR ranges * ⓘ

10.20.1.0/24

Source port ranges * ⓘ

*

Destination ⓘ

Any



Service ⓘ

Custom



Destination port ranges * ⓘ

8080

Protocol

☐ Any

☒ TCP

☐ UDP

☐ ICMPv4

☐ ICMPv6

Action

☒ Allow

☐ Deny

Figure 10. Rule tightened: Source 10.20.1.0/24, Destination port 8080, Protocol TCP.

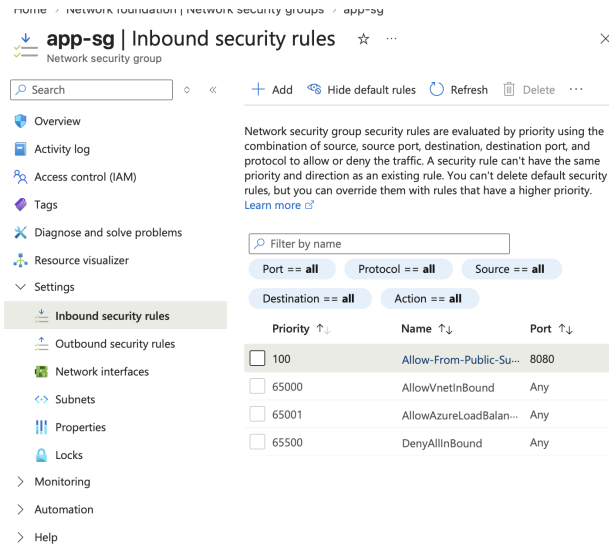


Figure 11. app-sg after: priority 100 rule now shows Port 8080, confirming the tightened rule saved.

data-sg

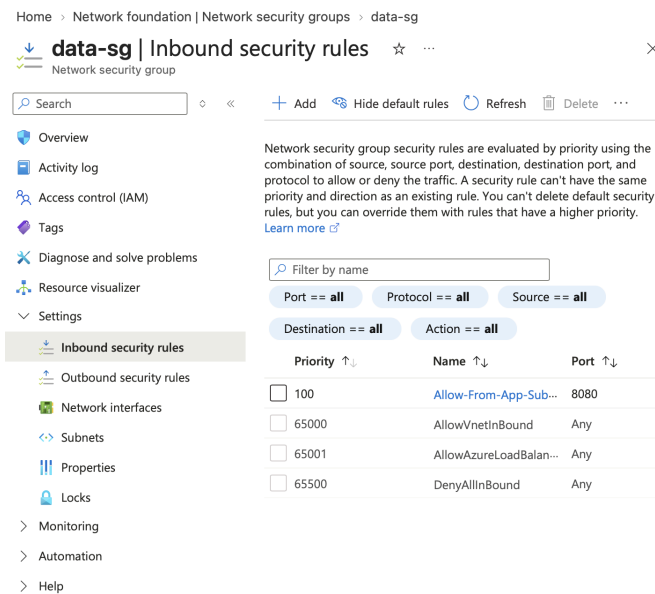


Figure 12. data-sg before: existing Allow-From-App-Subnet rule at port 8080 — reused the app tier's own port instead of a database port.

Home > Network for

data-sg

Network security

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve

Resource visualizer

Settings

Inbound security

Outbound security

Network interfaces

Subnets

Properties

Locks

Monitoring

Automation

Help

Allow-From-App-Subnet

data-sg

×

Source ⓘ

IP Addresses

▼

Source IP addresses/CIDR ranges * ⓘ

10.20.2.0/24

Source port ranges * ⓘ

*

Destination ⓘ

Any

▼

Service ⓘ

Custom

▼

Destination port ranges * ⓘ

5432

✓

Protocol

☐ Any

☒ TCP

☐ UDP

☐ ICMPv4

☐ ICMPv6

Action

Figure 13. Rule corrected: Source 10.20.2.0/24, Destination port 5432 (PostgreSQL), Protocol TCP.

7. Private Resource Validation — No Public IP

The rubric asks for hands-on proof that a security pattern actually works, not just configuration in isolation. The plan was to deploy a small test VM into snet-app with no public IP and connect to it through Bastion. In practice, the Azure subscription used for this course has no available vCPU quota for any small VM size (B-series, A-series, and even the smallest general-purpose sizes all returned "not available" for this subscription/region).

In plain terms — obstacle encountered — VM quota blocked, and a near-miss on cost

Every small VM size (B1s, B1ls, B2s, A1_v2) returned "not available" for this subscription in East US, confirmed by checking Azure's own Quotas page. While troubleshooting a categorized size list, a 368-vCPU HPC size (Standard_HB368-48rs_v5, priced at \$14,454/month) was briefly selected by mistake before being caught and reverted — flagged here deliberately as a real risk in cloud cost management, not hidden. Rather than keep fighting VM quota, the private-resource proof was rebuilt using an Azure Container Instance (aci-app-test) deployed with Networking type: Private into snet-app. Container Instances draw from separate quota and deploy in under a minute. This is a legitimate substitution — it demonstrates the same core control (a compute resource reachable only on the internal network, with no public IP) — but it means there is no literal "connected via Bastion" session screenshot for this specific resource, since Bastion's native RDP/SSH brokering targets VMs, not containers. Bastion itself is fully deployed and operational (Section 4); this note exists so the substitution is transparent rather than implied.

Create a virtual machine ...



Basics

Disks

Networking

Management

Monitoring

Advanced

Tags

...

☒ x64

Size * ⓘ

Standard_HB368-48rs_v5 - 368 vcpus, 448 GiB memory (\$14,454.00/month)

[See all sizes](#) | [Customize cores](#)

Run with Azure Spot discount ⓘ

☐

Administrator account

Authentication type * ⓘ

☒ SSH public key

☐ Password

ⓘ Azure now automatically generates an SSH key pair for you and allows you to store it for future use. It is a fast, simple, and secure way to connect to your virtual machine.

Username * ⓘ

azureuser

SSH public key source * ⓘ

Generate new key pair



Figure 14. Near-miss: Standard_HB368-48rs_v5 (\$14,454/month) briefly selected in the size picker before being caught.

Standard_A1

vCPUs : All

RAM (GiB) : All

Display cost : Monthly

Add filter

Showing 1 VM sizes. | Subscription: Azure subscription 1 | Region: East US | Current size: Standard_HB368-48rs_v5 | Image: Ubuntu Server 24.04 LTS
[Learn more about VM sizes](#)

> SKU	Type	vCPUs ↑	RAM (GiB)	Data disks	Ma
> Size not available	See the info bubble next to a size for details on availability				

Figure 15. Even the smallest general-purpose size (Standard_A1) returns "Size not available" for this subscription.

Create container instance ...

×

✓ Validation passed

BasicsNetworkingMonitoringAdvancedTagsReview + create

Basics

Subscription	Azure subscription 1
Resource group	rg-cloudsec1-can
Region	East US
Container name	aci-app-test
SKU	Standard
Image type	Public
Image	mcr.microsoft.com/azuredocs/aci-helloworld:latest
OS type	Linux
Memory (GiB)	1.5
Number of CPU cores	1
GPU type (preview)	None
GPU count	0

Networking

Networking type	Private
Ports	80 (TCP)
Virtual network	can-vnet-cloudsec1

Figure 16. Azure Container Instance created with Networking type: Private, attached to can-vnet-cloudsec1 / snet-app.

Home

Microsoft.ContainerInstances-20260928121533 | Overview

Deployment

Search x << Delete Cancel Redeploy Download Refresh

- Overview
- Inputs
- Outputs
- Template

✓ Your deployment is complete

Deployment name : Microsoft.ContainerInstances-20260928121533 Start time : 9/28/2026, 12:22:29 PM
Subscription : Azure subscription 1 Correlation ID : f7297dc4-0a23-42f6-8b63-4e2436f2d90e
Resource group : rg-cloudsec1-can

> Deployment details

✓ Next steps

[Go to resource](#)

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Figure 17. Deployment complete — aci-app-test, rg-cloudsec1-can, 9/28/2026.

aci-app-test

Container instances

Search ◇ << Start Restart Stop Delete Refresh ...

- Overview
- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Settings
- Monitoring
- Automation
- Help

⚠ Please be aware that Docker Hub has recently introduced a pull rate limit on Docker images. When specifying an image from the Docker Hub registry, this may impact your container instance.
[Learn more](#)

Essentials [JSON View](#)

Resource group (move)	: rg-cloudsec1-can
Status	: Running
Location	: East US
Subscription (move)	: Azure subscription 1
Subscription ID	: 9a8c964b-cf59-4657-bff2...
SKU	: Standard
OS type	: Linux
IP address (Private)	: 10.20.2.4
FQDN	: ---
Container count	: 1
Tags (edit)	: Add tags

Figure 18. aci-app-test Overview: Status Running, IP address (Private) 10.20.2.4, no public IP or FQDN listed anywhere in Essentials.

8. HIPAA & NIST CSF 2.0 Mapping

Consistent with the capstone's compliance-traceability requirement, both security patterns are mapped here to HIPAA Security Rule citations and NIST CSF 2.0 functions. This mapping is written to carry forward unchanged into the Week 9 Controls & Compliance Mapping deliverable.

Control	HIPAA citation	NIST CSF 2.0	How it's implemented
Bastion host (brokered remote access)	§164.312(a)(2)(i) — Unique user identification §164.312(e)(1) — Transmission security	PR.AC-3 Remote access is managed	Azure Bastion (Basic tier) is the sole path into private-tier resources; no direct public RDP/SSH exposure.
Zero-trust NSG segmentation	§164.312(a)(1) — Access control §164.312(e)(1) — Transmission security	PR.AC-5 Network integrity is protected (segmentation)	Explicit tier-to-tier allow rules (443→8080→5432); default-deny on everything else via NSG DenyAllInBound.
No public IP on private compute	§164.312(a)(1) — Access control	PR.AC-4 / PR.PT-4 Access permissions managed; communications protected	aci-app-test deployed with Networking type: Private — confirmed no public IP or FQDN on the resource.

9. Rubric Traceability

Rubric requirement	Status	Primary evidence
Three-tier diagram (public/private subnets, security groups, NACL-equivalent)	Complete	Figure 1
VPC/VNet configuration, subnet segmentation	Complete	Figures 2–4
Security group rules, documented configuration steps	Complete	Figures 9–13
Pattern 1 — Bastion host, config + screenshots	Complete	Figures 5–8
Pattern 2 — Zero-trust segmentation, config + screenshots	Complete	Figures 9–13
Proof of no public IP exposure on a private resource	Complete (ACI substitution documented)	Figures 16–18

10. Completion Statement

Criterion 2 — Cloud Architecture Design & Security Patterns — is complete. The environment demonstrates:

- A three-tier Azure VNet (can-vnet-cloudsec1, 10.20.0.0/16) with subnet-level segmentation across public, application, and data tiers, plus a dedicated Bastion subnet.
- Two working security design patterns: Azure Bastion (Basic tier) for brokered remote access, and zero-trust NSG micro-segmentation enforcing an explicit tier-to-tier allow chain.
- A private compute resource (aci-app-test) confirmed reachable only on the internal network, with no public IP or FQDN.
- An architecture diagram and full screenshot evidence, mapped to HIPAA Security Rule citations and NIST CSF 2.0 functions.

One adaptation is carried forward transparently rather than smoothed over: the private-resource proof uses an Azure Container Instance rather than a virtual machine, due to a subscription-level vCPU quota restriction encountered and documented in Section 7. This means there is no literal Bastion RDP/SSH session screenshot against this specific resource, though Bastion itself is fully deployed and operational. If the grading rubric requires that exact interaction, this is the one item to flag with the instructor ahead of submission.

Prepared for the Care Access Navigator capstone, Cloud Security 1 course track. Cross-references: rg-cloudsec1-can (Azure resource group), Criterion 1 deliverables (Security Concept Summary, Cloud Risk Assessment) already delivered under this same rubric.